

Public data

Biological holdout

Matched falsification

Qualified claim

Frozen human test

ST002081
1,539 samples · 112 people

Complete participant
holdout

Lipid families vs
size-matched random groups

Predictive compression
≠ family specificity

Locked human replication

ST000818
450 people · 15 populations

Complete population-
category holdout

Chemical descriptors vs
fixed-degree graph nulls

Replicated structure-aware
assay representation

Reaction-topology test

CCLE + HumanGEM
913 lines · 60 targets

Complete cancer-lineage
holdout

Direct neighborhoods vs
property-matched features

Compact topology signal;
no general interaction gain

Specificity requires out-of-group prediction and superiority to an equally difficult null.